Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Ctrl + Me

Project / Solution Name

SupplyPilot AI

Name (Leader)

WANG MINMIN

Email & Phone (Leader)

SWE2309536@xmu.edu.my | +60147149520

University Affiliation(s)

Xiamen University Malaysia

Live Prototype URL

https://supply-pilot-ai.replit.app/

Code Repository URL (public / access granted to judges)

https://github.com/DouHuaHua-Cathy/SupplyPilot-AI

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The publicly deployed app consists of a React/TypeScript interface and an Express backend served from the same Replit Autoscale web artifact. Staff enter a synthetic supplier confirmation for PO-1024; the API validates the request and calls OpenAI gpt-5-mini through Replit AI Integrations. The model extracts structured product quantities, availability, dates and quoted evidence. Zod validates the response, while a further check ensures evidence appears in the input. Separate deterministic TypeScript rules compare the facts with the purchase order, flag issues and prepare a follow-up draft.

Staff can review/edit the draft; a simulated Owner can approve or reject it. Activity history and drafts persist in that browser's localStorage only. No message is sent to a supplier; no payments or real customer records are processed. If the AI service is unavailable, a clearly labeled rules-only fallback remains usable. Usage controls include per-IP rate limiting, a process-local daily request cap and duplicate-request reuse. There is no production database or live Orbyte API connection. The architecture could later expose a reusable Orbyte procurement connector with server-side identity and audit logs.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
pnpm (workspace)	Dependencies / lockfile	MIT
TypeScript (lockfile)	Frontend, backend and rules	Apache-2.0
React / ReactDOM (lockfile)	Interactive UI	MIT
Express / Node.js (lockfile)	HTTP API and runtime	MIT
Vite (lockfile)	Frontend build	MIT
Zod / Tailwind CSS (lockfile)	Validation / styling	MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Replit AI Integrations (OpenAI)	Production supplier-fact extraction	Replit-managed / metered
OpenRouter (historical test only)	Earlier live API integration tests; NOT deployed	Free experimental
Google Gemini API (test only)	Earlier model/SDK experiments; NOT deployed	API key / dev testing
No messaging/payments APIs	Messages are drafted and copied, never sent	None

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale	Live React + Express hosting	Autoscale / Asia
Replit Workspace	Development, previews, Agent	Core account
Replit AI Integrations	Managed GenAI API access	Metered via Replit
GitHub	Public source / commit history	Public repository
Database	No server DB; browser localStorage only	None (demo)

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-5-mini	OpenAI via Replit	Pre-trained / hosted; production	OpenAI API terms
None (fine-tuned)	Not applicable	No fine-tuning or training	N/A
Gemini / OpenRouter models	Direct API tests only	Exploratory; NOT current runtime	Provider terms

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
PO-1024 + supplier notes	Team-generated	Synthetic demo data	Team-created	No
Validation cases	Team-generated	Synthetic test fixtures	Team-created	No
Real SME data	None	Not collected or used	N/A	No

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit Agent	Application coding, testing, deploy setup	Major
OpenAI Codex	Local feature implementation, tests and debugging	Major
ChatGPT	Planning, prompts, documentation, QA support	Major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated changes were checked against the purchase-order requirements, original-message evidence and deterministic calculation rules. The team tested complete, delayed, short, unavailable and incomplete supplier responses; wrote automated frontend/API tests; ran TypeScript checks and production builds; and exercised the deployed Replit app. Unexpected or unsupported model results are rejected or handled by a clearly marked rules-only fallback. Final supplier communication always requires human review. The team remains responsible for the code, content, limitations and any AI-assisted development decisions.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Agent workspace; AI Integrations	Built, tested and served live app
Replit	Autoscale / cloud hosting	Public full-stack demo
RunPod	Not used (N/A)	No self-managed GPU work

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was central to the working MVP: Replit Agent helped develop and test the application; Replit AI Integrations supplies managed OpenAI model access; and Replit Autoscale hosts the public full-stack app. The technical workflow could be migrated to another host/model gateway, but doing so would require new integration and deployment validation. RunPod was not used because the team did not train, fine-tune or self-host an inference model on a GPU.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The public demonstration instructs visitors to enter only synthetic supplier messages. There is no signup, production customer database or server-side user profile. Supplier message text entered in AI mode is sent from the browser to the Express API and then to Replit-managed AI Integrations/OpenAI for extraction; provider-side processing/retention follows the applicable service terms, which the team does not control. Drafts, simulated roles, approvals and activity records are stored in browser localStorage and persist until cleared by that browser/user. The backend receives messages transiently, and the product does not intentionally store raw message text in a database. API credentials remain server-side in Replit deployment secrets; they are not distributed in client code. The demo is not appropriate for sensitive commercial data or real personal records. A production release would require user consent notices, a data-retention policy, authenticated roles and centralized audit storage.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Model outputs can omit details, misread ambiguous dates/units or provide unsupported evidence. The API applies structured validation and an evidence-presence check, and deterministic TypeScript - rather than the model - computes purchase-order differences. Uncertain or absent confirmations are shown separately. Model latency, provider outages and quotas can make AI unavailable; the UI exposes the failure and falls back to labeled rules-only analysis. AI request caps are process-local and reset after restart; they do not guarantee a fixed credit bill. Staff/Owner selection is simulated in the browser, not true authorization. Activity history lives in localStorage, not a tamper-resistant server audit log. A separate dependency audit flagged four high-severity advisories; updating and retesting the dependencies remains necessary. The synthetic single-order demo has limited generalizability; multi-order testing, real SME usability measurement and security hardening remain Phase 2 work.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

24 October 2026